

Prompt for Transformation:

Please translate the Natural Language into STL specification.

Let a and b be two variables, and let φ be the specification.

The rules are as follows:

- $\varphi_1 U[a,b] \varphi_2$ indicates that there exists a moment t' such that φ_1 is satisfied before t', and φ_2 is satisfied at t', where t' is within a time distance of a to b from the current moment.
- $F[a,b] \varphi$ indicates that there exists a point within the interval [a, b] where φ is satisfied.
- $G[a,b] \varphi$ indicates that φ is satisfied at every point within the interval [a, b].

Additionally, assume signals $x_1[t]$, $x_2[t]$, . . . , $x_n[t]$, then atomic predicates are of the form: $f(x_1[t], . . . , x_n[t]) > 0$.

The STL formula should only contain atomic propositions, boolean operators $\wedge, \vee, \neg, \rightarrow, \leftrightarrow$ and temporal operators $U[a,b], G[a,b], F[a,b]$

Output the STL specification:

Prompt for Refine the Ambiguity/Incompleteness Requirement:

Your task is to refine the original natural language requirement to resolve ambiguity.

Original requirement: [input requirement]

Reason for ambiguity: [query of ambiguity]

Clarifying sentence for this ambiguity: [user feedback]

Please integrate the information from the clarifying sentence into the original requirement seamlessly, guided by the reason for the ambiguity.

Output only the refined requirement:

Prompt for Back-Translation:

You are an expert in Signal Temporal Logic (STL) translation. Your task is to convert any given STL formula into natural language, following these rules:

•{prop} represents an atomic proposition.

Mapping rules:

- $\neg\{prop\} \rightarrow$ "not {prop}"
- $\{prop_1\} \wedge \{prop_2\} \rightarrow$ "{prop_1} and {prop_2}"
- $\{prop_1\} \vee \{prop_2\} \rightarrow$ "{prop_1} or {prop_2}"
- $\{prop_1\} \rightarrow \{prop_2\} \rightarrow$ "if {prop_1}, then {prop_2}"
- $\{prop_1\} \leftrightarrow \{prop_2\} \rightarrow$ "{prop_1} if and only if {prop_2}"
- $G[L,U] \{prop\} \rightarrow$ "for the next L to U time units, always {prop}"
- $F[L,U] \{prop\} \rightarrow$ "for the next L to U time units, eventually {prop} holds"
- $\{prop_1\} U [L,U] \{prop_2\} \rightarrow$ "for the next L to U time units, {prop_1} holds continuously until {prop_2} holds"

Process:

- 1.Decompose the STL formula into atomic propositions and operators.
- 2.Assign placeholders {prop_1}, {prop_2}, etc., to subformulas.
- 3.Translate atomic propositions first, then combine placeholders according to Boolean and temporal operators.
- 4.Maintain logical and temporal order throughout the process.
- 5.Output the final natural language.

Prompt for Refine the Incomplete Requirement:

Your task is to refine the original natural language requirement to resolve incompleteness.

Original requirement: [input requirement]

Reason for incompleteness: [query of incompleteness]

Clarifying sentence for this incompleteness: [user feedback]

Please seamlessly integrate the information from the clarifying sentence into the original requirement, guided by the reason for incompleteness.

Output only the refined requirement:

Prompt for Few-Shot Incompleteness/Ambiguity Detection :

Your task is to determine whether a given natural language requirement contains incomplete/ambiguous information.

Follow the examples below. If the requirement contains no incomplete/ambiguous information, output No. If it does contain such information, output Yes.

Examples:

<input requirement, detection result>

Output only the detection result:

Prompt for Few-Shot Incompleteness/Ambiguity Query Generation :

Your task is to generate clarification queries for incomplete /ambiguous natural language requirements.

You should follow the examples below, analyze the reason why a requirement is incomplete/ambiguous, and then output an appropriate query.

Examples:

<incomplete/ambiguous requirement, reason for incompleteness/ambiguity, query>

Output only the query: